

1. Immutable Edge Deployments for Passenger Engagement

- **Aligned Objective:** Delivering Software-Defined Passenger Engagement (Edge Computing) in disconnected environments 1, 2.
- **Influence on Success Metrics:** Reduces the time required for FAA compliance sign-offs and virtually eliminates deployment downtime, as updates that go wrong can be immediately rolled back to the previous version like a smartphone 3, 4.
- **Red Hat Technology:** RHEL 10 Image Mode and Red Hat Device Edge, treating the OS as an immutable container 2, 5, 6.
- **RH IT Partner:** AWS (acting as the cloud backend for staging image builds before pushing them to the aircraft edge) 1, 2.

2. VM to Container Lift-and-Shift Strategy

- **Aligned Objective:** Migrating off legacy virtualization to escape ballooning licensing costs following the Broadcom acquisition 7.
- **Influence on Success Metrics:** Drives rapid reduction in infrastructure licensing costs by avoiding duplication, transferring existing VM workloads seamlessly to modern cloud or container environments 8.
- **Red Hat Technology:** Red Hat OpenShift Virtualization, which allows running both VMs and containers side-by-side on a single unified platform 9.
- **RH IT Partner:** Microsoft (leveraging Azure as the deployment destination) 8, 10.

3. Zero-Downtime Event-Driven MRO (Maintenance, Repair, Overhaul)

- **Aligned Objective:** Automating IT service management and transitioning to proactive maintenance 2, 11, 12.
- **Influence on Success Metrics:** Minimizes aircraft downtime and increases engineering velocity by eliminating the manual polling of service desks, running automation asynchronously the moment an approval is granted 13, 14.
- **Red Hat Technology:** Event-Driven Ansible (EDA) 11, 15.
- **RH IT Partner:** ServiceNow (via the ServiceNow ITSM event source plugin) 16, 17.

4. Legacy Java App Containerization for Cybersecurity

- **Aligned Objective:** Overcoming massive technical debt, specifically the retirement of CentOS 7 and aging Java applications 18, 19.
- **Influence on Success Metrics:** Dramatically improves internal cybersecurity scores (driving toward 100%) and guarantees the retention of security waivers by isolating vulnerable apps 19, 20.
- **Red Hat Technology:** Red Hat OpenShift and RHEL Universal Base Image (UBI) as a starting point for secure, hardened containers 21, 22.
- **RH IT Partner:** CrowdStrike (to validate the container images and issue security compliance sign-offs) 23, 24.

5. Infrastructure as Code (IaC) Standardization

- **Aligned Objective:** Standardizing a historically fragmented engineering culture where developers "rolled their own stuff" without controls 25.
- **Influence on Success Metrics:** Accelerates provisioning times and eliminates configuration drift across thousands of global host machines 26, 27.
- **Red Hat Technology:** Ansible Automation Platform (AAP 2.6) integrated directly with Terraform workflows 28.

- **RH IT Partner:** HashiCorp (creators of Terraform and Packer) 27-29.

6. AI-Assisted Administration for Aviation Systems

- **Aligned Objective:** Cloud Migration and addressing the technical skill gaps within the engineering teams 30.
- **Influence on Success Metrics:** Decreases the time-to-resolution for internal troubleshooting, accelerates script generation, and shortens the onboarding time for new Linux administrators 30, 31.
- **Red Hat Technology:** RHEL 10 Command Line Assist and Red Hat Lightspeed 31-33.
- **RH IT Partner:** SHI (as the Value-Added Reseller managing the procurement and licensing distribution of these advanced AI features) 34.

7. Automated Air-Gapped Media Content Delivery

- **Aligned Objective:** Modernizing the highly locked-down, almost air-gapped Media Content Service environment 35, 36.
- **Influence on Success Metrics:** Increases the frequency and reliability of inflight media content updates while maintaining absolute security in disconnected environments 36, 37.
- **Red Hat Technology:** Ansible Automation Platform combined with Red Hat Satellite (for importing necessary data and updates into a local, isolated model) 37.
- **RH IT Partner:** GitLab (to securely source control the automated deployment pipelines) 38-40.

8. Quantum-Safe LEO Connectivity Infrastructure

- **Aligned Objective:** Securing the massive amounts of data flowing through new Multi-Orbit and Low Earth Orbit (LEO) satellite networks 1, 12.
- **Influence on Success Metrics:** Ensures long-term regulatory compliance and a zero-breach data posture against future cryptographic threats, crucial for handling airline passenger payment data 5, 41.
- **Red Hat Technology:** RHEL 10 featuring out-of-the-box, FIPS-compliant quantum cryptography enabled at install time 41, 42.
- **RH IT Partner:** CrowdStrike (to monitor and validate the quantum-safe encryption layers across the fleet) 23, 24.

9. Dynamic Network Inventory Synchronization

- **Aligned Objective:** Establishing a single "source of truth" to effectively map and automate the sprawling global network of physical and virtual devices 43, 44.
- **Influence on Success Metrics:** Enhances Configuration Management Database (CMDB) accuracy and drastically reduces duplicate Configuration Items (CIs) in the IT service desk 45, 46.
- **Red Hat Technology:** Ansible Automation Platform utilizing the NetBox dynamic inventory plugin 38, 43, 47.
- **RH IT Partner:** ServiceNow (to receive the organically managed, deduplicated CI updates from Ansible) 46, 47.

10. Multi-Cloud CI/CD Portability Hub

- **Aligned Objective:** Shifting from a hardware OEM to a software-and-services (SaaS/PaaS) provider without vendor lock-in 1, 48.

- **Influence on Success Metrics:** Optimizes cloud infrastructure costs and provides the flexibility to migrate workloads effortlessly between Azure, AWS, and on-premises environments 8, 9, 28.
- **Red Hat Technology:** Red Hat OpenShift, providing a uniform, agnostic platform for containers regardless of the underlying hardware or cloud provider 9, 49.
- **RH IT Partner:** Microsoft Azure (as the primary initial cloud destination for these containerized workloads) 10, 50.